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Semantics-aware location privacy preserving: A differential privacy approach

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ABSTRACT

The protection of location privacy, as a highly sensitive characteristic of information, has been extensively analyzed and discussed for a significant period. Recently, exploiting the semantics of locations offers a new dimension to enhance privacy preservation by enabling more effective control over the information disclosed by users. Different from most prior research efforts, which regard location semantics as a category, in this paper, location semantics is the statistical information about the Points of Interest (PoIs) in the specific location's vicinity, which can be represented as a multi-dimensional vector. Further, Semantic Indistinguishability (Sem-Ind), a more relaxed privacy guarantee for location privacy than Geo-Indistinguishability (Geo-Ind), is derived under the paradigm of differential privacy. Multiple location obfuscation mechanisms, which integrate linear programming and heuristic search, respectively, are proposed to reduce utility loss while ensuring Sem-Ind. Based on the defined utility and privacy metrics, these obfuscation mechanisms are empirically evaluated on the GeoLife dataset. Experimental results indicate that the existing Geo-Ind-based obfuscation mechanisms satisfy Sem-Ind at an excessive loss of utility. Furthermore, the linear programming-based approach is capable of discovering optimal obfuscation functions, whereas the heuristic algorithms are more efficient in obtaining acceptable utility results.

1. Introduction

The development of mobile communication technologies has led to a significant increase in the quantity and diversity of mobile applications available to individuals in their social and personal lives. Among these, location-based services (LBSs) form a prominent category, requiring users' geographic location information to deliver personalized and efficient services. To utilize location-based services, users need to submit their geographic location data to service providers. However, location data itself often contains substantial sensitive information about users. Additional sensitive attributes can be derived from these location traces, including users' behavioral patterns, social relationships, and personal preferences. The acquisition of such data by malicious third parties or its utilization by service providers with ill intentions gives rise to user privacy breaches.

Fortunately, numerous researchers have acknowledged the significance of location privacy, leading to the proposal of Location Privacy Protection Mechanisms (LPPMs) to reduce the extent of location privacy breaches (Primault et al., 2019; Jiang et al., 2021; Kim et al., 2021). Obfuscation is one of the most commonly used methods for protecting location privacy. In this process, users' actual locations are replaced with pseudo-locations before being uploaded. Numerous privacy concepts are employed in the development and evaluation of privacy preserving mechanisms using pseudo-locations, including K-anonymity-based (Konstantinidis et al., 2015; Zhao et al., 2018), differential privacy (DP)-based (Yu et al., 2017; Cao et al., 2020a), Bayesian inference-based (Yu et al., 2017; Zhang et al., 2022), and cryptography-based (Jiang et al., 2019) approaches. One of the most well-known location privacy preservation mechanisms based on differential privacy is Geo-Indistinguishability (Geo-Ind), replacing the user's true location x with a pseudo-location z. Given a set of candidate locations, an adversary cannot distinguish the probability of any location being the user's true location from this candidate set.

Prior location obfuscation mechanisms primarily focus on the privacy implications of revealing pseudo-locations. However, the situation can change significantly when the focus shifts from locations to the observation of location semantics. Semantic features associated with locations have the potential to reveal sensitive attributes of both the locations and the users, and can be exploited by malicious actors (Cao et al., 2018). Existing location obfuscation mechanisms aim to

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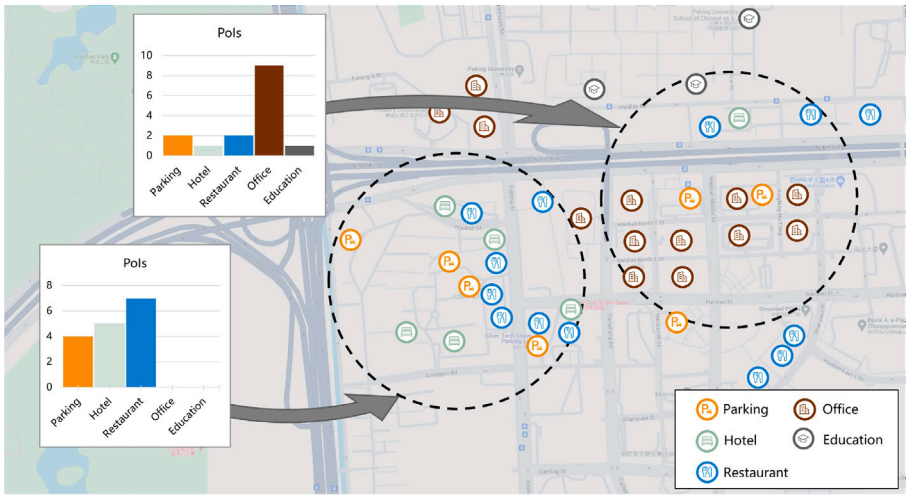

Fig. 1. Illustration of nearby PoIs query of two locations in the map.

prevent the leakage of users' private location presence through pseudo-locations. Nevertheless, they may fall short in addressing the private information leaked by location semantics or may impose overly strict privacy constraints, resulting in substantial utility loss.

When discussing location semantics, various definitions exist in the research literature, such as location categories, place intensity, and more. In this paper, location semantics is defined as the aggregate of distinct types of places, such as transportation infrastructure and residential buildings, aka Points of Interests (PoIs), within a certain radius from the location centroid. Based on the number of specified categories of PoIs, a histogram can be constructed to partially capture the characteristics of location semantics. This approach provides a novel perspective for measuring a user's location privacy based on the distribution of PoIs surrounding a location.

For example, in Fig. 1, two locations revealed by the user have different PoI distributions within the range of 250 m. More parking lots, hotels and restaurants are distributed near the left location, while there are more office buildings near the right one. In this paper, the semantics of the two locations can be denoted as the distribution of various types of PoIs, which are $y_{left} = [4, 5, 7, 0, 0]$ and $y_{right} = [2, 1, 2, 9, 1]$. For the one-dimensional scenario, the semantics is a scalar. In the following sections of this paper, we will examine both cases where the semantic representation is either a single-dimensional or multi-dimensional vector.

Location semantics, as characterized by PoI distributions, inherently reflect location attributes. An adversary with access to historical records of users' true locations can extract sensitive personal information from these attributes, including income status, gender, and educational background. For instance, frequent presence and stays at locations with high densities of educational institutions strongly indicate the user's identity as a student or faculty. On the other hand, as stated in Tong et al. (2021), an adversary can infer user locations by obtaining multiple category PoI distributions, leading to the disclosure of location information. Consequently, to reduce privacy breaches, it becomes crucial to prevent adversaries from obtaining the true location semantics of user visits.

In this paper, we define semantic location privacy termed Semantic Indistinguishability (Sem-Ind), under the paradigm of differential privacy. Sem-Ind limits the disclosure of semantic information by imposing a theoretical constraint on the probability of location obfuscation, which is similar to other privacy guarantees like Geo-Ind. More precisely, all locations on the map are partitioned into multiple semantic equivalent sets based on their semantic values. The user's true location will be obfuscated to any location within the corresponding equivalence set. As will be rigorously defined in Section 4.3, the probabilities of all locations being obfuscated to the set should be indistinguishable from each other. Compared with other location privacy notions based on differential privacy, Sem-Ind emphasizes the semantics of pseudo-locations after obfuscation, applying constraints on the obfuscation probabilities that are more relaxed than those in other approaches.

Under the constraints of Sem-Ind, obfuscation mechanisms should be designed to balance privacy gain and utility loss, and be evaluated by their metrics respectively. The paper proposes several privacy-preserving location obfuscation mechanisms based on treating location semantics as discrete numerical vectors with a specified value range determined by the city morphology. It is clear that while Geo-Ind-based mechanisms also provide Sem-Ind guarantee, they lead to excessive utility loss. To further improve the utility, we formulate our challenge as a linear programming problem, exploring the boundary of Sem-Ind constraints beyond Geo-Ind. As to computing efficiency, two heuristic algorithms are proposed, utilizing the post-processing properties of differential privacy to achieve acceptable utility loss while maintaining semantic location privacy. These algorithms are easy to comprehend and construct, offering a novel viewpoint on the trade-off between privacy and utility. Our contributions are as follows:

- To the best of our knowledge, it is the first time to take multi-categorical nearby PoI distributions as location semantics. Sem-Ind is proposed under the paradigm of differential privacy, requiring that the probabilities of obfuscating locations to pseudo-locations in semantic equivalent sets should be indistinguishable.
- Taking into account the privacy and utility metrics, we devise three obfuscation mechanisms including the adaptation of Geo-Ind, linear programming and heuristic mechanism, all of which are constructed to minimize utility loss under the constraints of Sem-Ind.
- Finally, using the GeoLife dataset, we perform an empirical evaluation for all the mechanisms. The results indicate that the linear programming-based algorithm serves as the baseline of utility loss, whereas heuristic mechanisms perform efficiently. Both of them are superior to the adaptation of Geo-Ind in the context of Sem-Ind. In summary, Sem-Ind is a fairly relaxed privacy guarantee that does not necessitate as substantial a utility loss as other privacy concepts.

The remainder of this paper is organized as follows. The related literature is discussed in Section 2. Section 3 introduces some necessary background and Section 4 formulates the definition of semantic location privacy as well as the privacy protection problem. The mechanisms we propose and analyze are described in Section 5. The results of evaluations are given in Section 6. Finally, we conclude the paper in Section 9.

2. Related work

2.1. Location privacy preserving mechanism

As LBS applications have expanded in recent years, location privacy has become an increasingly prominent concern, leading to a considerable amount of research focused on generic privacy-preserving mechanisms (Primault et al., 2019; Jiang et al., 2021), as well as privacy constraints specific to particular scenarios in applications (Kim et al., 2022). According to the taxonomy proposed by Vincent et al. LPPMs can be mainly divided into four categories including cloaking (generalization) (Xiao et al., 2017; Huguenin et al., 2018), dummy-based (Xu et al., 2021; Wu et al., 2020), protocol-based (Aïvodji et al., 2018; Jiang et al., 2019) and obfuscation (perturbation) (Andrés et al., 2013). Among them, obfuscation under the differential privacy (DP) (Dwork and Roth, 2014) guarantee which is adopted from database privacy has been widely analyzed and applied to location privacy protection. Andrés et al. proposed Geo-Indistinguishability (Geo-Ind), a location privacy guarantee based on local DP that places rigorous theoretical constraints on the probability distributions of location obfuscation algorithms (Andrés et al., 2013). For any two locations that can be obfuscated to the same pseudo-location, the ratio of their obfuscation probabilities should be proportional to the distance between them.

Despite the rigorous privacy guarantees provided by Geo-Ind, the Planar Laplacian mechanism used to obfuscate locations does not take utility loss into account. Based on this observation, in Bordenabe et al. (2014), considering the implication of privacy on utility which is measured by expected obfuscation error, Bordenabe et al. formulated the tradeoff between these two sides as a linear programming problem. To improve the computational efficiency of the optimization process, a spanner graph-based approximation is introduced to reduce the number of constraints. Furthermore, in Chatzikokolakis et al. (2017), a more efficient post-processing method based on Bayesian remapping was proposed as an alternative to linear programming. Note that Geo-Ind only limits the disclosure of information resulting from observations, which means it may be vulnerable to Bayesian adversaries equipped with prior knowledge. To mitigate this issue, Yu et al. proposed a dynamic obfuscation approach called PIVE, which jointly accounts for the lower bounds of inference error and DP guarantee (Yu et al., 2017). PIVE first generates the protection location set which provides enough magnitude of inference error given the specified prior, then samples pseudo-locations to ensure DP over the set. Besides the generic LPPM, Geo-Ind has also been introduced into different application scenarios as a basic privacy protection requirement (Wang et al., 2020; Tao et al., 2020; Liu et al., 2023; Wang et al., 2021). In all, these works focus on obfuscating actual locations to pseudo-locations, through which extra knowledge cannot be obtained by the adversary.

2.2. Location semantics privacy

Besides the user's true location, location semantics reflecting the location properties are regarded as indirect yet sensitive sources of personal information susceptible to extraction and analysis. However, unlike the clear-cut definition of locations, the term "semantics" can refer to a variety of meanings, ranging from place categories, user stay points to crowd densities. Several preliminary works have investigated privacy protection mechanisms for location semantic information.

One way to represent location semantics is to use semantic tags, like restaurants, hospitals, railway stations and others. In Bostanipour and Theodorakopoulos (2020) and Qiu et al. (2021), the semantic relationships between locations are organized as a hierarchical tree structure, where abstract semantic categories (e.g., 'restaurant') serve as parent nodes, while more specific semantics (e.g., 'Chinese restaurant', 'Western restaurant') are represented as child nodes. Then the detailed semantic information of locations is abstracted to a higher-level semantic category, thereby concealing specific semantic details within more general classifications. User mobility can be modeled as a Dynamic Bayesian Network that considers locations and semantic tags cooperatively, as proposed by Bostanipour et al. in Bostanipour and Theodorakopoulos (2020). They also introduced an inference attack and formulated a joint obfuscation approach to bound the adversary's inference error to ensure privacy.

To the best of our knowledge, the only research that takes location semantics as distributions of multiple categories nearby PoIs is Tu et al. (2017). One of the two attack models raised by Tu et al. was named semantic attack, by which the attacker could obtain the location's semantic information, i.e., its main function of socioeconomic activities, through comparing nearby PoIs distribution of the location with that of the whole city. Consequently, the attacker can infer the individual's motivation for visiting these locations, leading to a breach of privacy. Then, Tu et al. incorporated the definition of location semantics into the privacy model under the paradigm of k-anonymity, l-diversity and t-closeness. This paper follows a definition of location semantics similar to Tu et al. (2017). However, the privacy guarantee is established under the paradigm of differential privacy.

Some other works also considered the privacy issue of disclosing nearby PoIs, since the PoIs distribution can be regarded as a multi-dimensional feature describing the factors of both the urban morphology and the individual's behavior pattern. Cao et al. evaluated the uniqueness of PoI distributions over the city, revealing nearby PoIs can lead to privacy breaches called location re-identification. Therefore, the disclosure of PoI information should be carefully designed and analyzed to protect location privacy (Cao et al., 2018). Based on the evaluation of PoI uniqueness, a more practical relocation attack together with a secure PoI release mechanism was proposed by Tong et al. (2021). A straightforward privacy enhancement algorithm was proposed, which added noise to the PoIs distribution to provide a differential privacy guarantee. However, the utility of the LBS application, which can be reflected as the location distortion, was ignored by Tong et al. (2021). Furthermore, injecting noise directly into the PoI distribution might result in distributions that do not exist in any real-world locations. These non-existent distributions would be meaningless from both the perspective of LBS utility and the privacy protection against adversaries.

More advanced attack vectors have been proposed that rely on the numerical features associated with PoIs, revealing significant potential knowledge that can be extracted. In Cao et al. (2020b), a representative learning-based method, called Habit2vec was proposed to extract semantics from the PoIs in trajectory data. Trajectory semantics were represented as multi-dimension vectors that preserved personal habit information. Wu et al. proposed a machine learning-based epidemiological investigation method that involves classification and clustering, to extract personal information from trajectory data including PoIs (Wu et al., 2021). A rising number of research are examining the issue of location privacy breaches caused by PoIs. However, so far, there remains a gap in the literature regarding the release of PoI data through location obfuscation under the framework of DP.

3. Preliminaries

3.1. Local differential privacy

Differential privacy (Dwork and Roth, 2014), regarded as a de-facto standard, is a rigorous paradigm that is commonly used in database management systems by data curators, providing a provable privacy guarantee for individual data. In DP, the query result should be partially randomized by data curators so that adversaries cannot glean information about specific individuals within the protected data set. One common approach is to add noise sampled from a specified distribution, such as Laplacian or Gaussian distribution (McSherry and Talwar, 2007) to provide a DP guarantee.

Recently, local differential privacy (LDP) (Yang et al., 2020) has been proposed as a new DP model, which reduces reliance on the trust-

worthiness of data curators. Obfuscation is required to be performed on the client-side, instead of the data curator's side. LDP requires that the probability of any input value being obfuscated into a given output value through the randomization mechanism should be approximately equal for all possible input values.

Definition 1 (*ϵ-Local-Differential-Privacy*)**.** Let $\mathcal{X}$ be the domain of input value and $\mathcal{S}$ the domain of obfuscated value, a randomized mechanism $\mathcal{M}$ provides ϵ-local-differential-privacy if for all $x_1, x_2 \in \mathcal{X}$ and any potential output value $s \in \mathcal{S}$,

$$\frac{\Pr[\mathcal{M}(x_1) = s]}{\Pr[\mathcal{M}(x_2) = s]} \leq e^{\epsilon}. \tag{1}$$

A fundamental characteristic of differential privacy is its resilience to post-processing. When any data-independent function is added after the obfuscation mechanism $\mathcal{M}$, the resulting composite mechanism still satisfies DP requirements.

Theorem 1 (*Post-Processing*)**.** *Let $\mathcal{M}$ be an ϵ-differential private mechanism and g be an arbitrary mapping from the set of possible outputs to an arbitrary set. Then, $g \circ \mathcal{M}$ is ϵ-differentially private.*

3.2. Location obfuscation mechanism

To protect users' location privacy, true locations are substituted with fake ones, which will be revealed to LBS and adversaries. Such a method is called location obfuscation, and the substituted locations are typically referred to as pseudo-locations. Formally, an obfuscation mechanism $\mathcal{M}$ will take the true location x as the input, then output the pseudo-location z, either deterministically or randomly. An obfuscation mechanism $\mathcal{M}$ defines the conditional probability distribution $f(.|.)$ of generating pseudo-location O_l given the user's true location A:

$$f(z|x) = \Pr[O_l = z | A = x]. \tag{2}$$

In this paper, the entire region under consideration is divided into discrete locations forming a grid structure, which is represented by the location set $\mathcal{X}$.

3.3. Geo-indistinguishability

The most well-known and commonly used location obfuscation mechanism is Geo-Indistinguishability (Andrés et al., 2013) built on $d_{\mathcal{X}}$-privacy in the framework of LDP. Given any two locations x_1 and x_2, the difference in the conditional probability distributions of reporting an arbitrary z should be exponentially proportional to the geographic distance between x_1 and x_2. The logarithm of the ratio between these two probabilities relaxes proportionally as the distance between locations increases. Thus, locations that are geographically close have similar probability distributions for their obfuscation.

Definition 2 (*ϵ-Geo-Indistinguishability*)**.** A mechanism $\mathcal{M} : \mathcal{X} \to \mathcal{X}$ provides ϵ-Geo-Indistinguishability if for all $x_1, x_2 \in \mathcal{X}$ and any potential pseudo-location $z \in \mathcal{X}$, where $\mathcal{X}$ is the set of all the locations in the map,

$$\frac{\Pr[\mathcal{M}(x_1) = z]}{\Pr[\mathcal{M}(x_2) = z]} \leq e^{\epsilon d(x_1, x_2)}, \tag{3}$$

where $d(x_1, x_2)$ is the geographic distance between x_1 and x_2.

One of the LPPMs guaranteeing Geo-Ind is the Planar Laplacian mechanism (Andrés et al., 2013). For a given location x, it first draws (θ, r) from the polar Laplacian distribution by sampling θ uniformly in $[0, 2\pi)$ and setting $r = C_\epsilon^{-1}(p)$ where p is uniformly sampled in $[0, 1)$, and

$$C_\epsilon^{-1}(p) = -\frac{1}{\epsilon}(W_{-1}(\frac{p-1}{e}) + 1), \tag{4}$$

in which W_{-1} is the Lambert W function (the -1 branch).

3.4. Bayesian adversary

A Bayesian adversary represents an attack model in privacy protection scenarios. This type of adversary possesses prior knowledge about the system and users, typically in the form of probability distributions, and employs Bayesian inference to update its knowledge based on observed data. Specifically, the adversary starts with a prior probability $\pi(x)$ about the target information x, observes the obfuscated data z, and uses Bayes' theorem to compute the posterior probability $Pr(x|z)$:

$$\Pr(x|z) = \frac{\pi(x) f(z|x)}{\sum_{x_i \in \mathcal{X}} \pi(x_i) f(z|x_i)}. \tag{5}$$

The adversary can perform an optimal inference attack to guess the actual location $\hat{x}$ using the posterior probabilities, where the objective is to minimize the expected inference error:

$$\hat{x} = argmin_{x \in \mathcal{X}} \sum_{x_i \in \mathcal{X}} \Pr(x_i|z) d(x, x_i). \tag{6}$$

4. Problem formulation

In this section, we present the details of the system containing the interaction between related participants, and subsequently outline the assumptions concerning the potential threats imposed by adversaries. Thereafter, we introduce the theoretical formulation of the privacy guarantee and proceed to discuss the quantitative evaluation metrics.

4.1. System overview

A diagram of the system model is shown in Fig. 2 which includes two entities: mobile clients and the location-based service provider (LBSP). Once the user requests a location service, the mobile clients will obtain the current location and apply location obfuscation to it. The service request containing the pseudo-location z is sent to the LBSP, usually through the mobile network. Then the LBSP will respond with the required service results according to the location z back to the clients. Note that we only consider the case of preserving the privacy of a single location. We leave the research considering multiple consecutive locations (trajectory) to future work.

In this paper we focus on reducing the private information leaked from the location semantics y when users submit their pseudo-location z. Both in practical applications and existing research literature, the semantics of a location typically include a range of meanings, such as places, road networks, crowd density, and more, as referenced in various studies (Li et al., 2019; Chatzikokolakis et al., 2015). Following the definition of location semantics proposed in Tu et al. (2017), in this paper, we define the semantics of a location as the distribution of PoIs within a certain range in the vicinity, as private individual information related to jobs, residences may be inferred from the pattern of the PoIs distribution near the locations the user visits.

Formally, suppose there are N categories of PoIs, and the count of PoIs which belong to category j is $y_j \in \mathbb{Z}$. Apparently, the semantic data considering all N categories of PoIs can be denoted as an N-dimensional integer vector $y \in \mathcal{Y} \subset \mathbb{Z}^N$, where $\mathcal{Y}$ is the domain of existing semantics in the scope $\mathcal{X}$. The relationship between a location x and its semantic y can be formulated as a deterministic query function $\mathcal{K}_r(x) : \mathcal{X} \to \mathcal{Y}$, where the parameter r denotes the radius of the nearby PoIs query. Note that $\mathcal{K}_r(x)$ is determined by urban morphology in the geographic region $\mathcal{X}$, and remains fixed within a period of time (Cao et al., 2018). For clarity, we will omit the parameter r in the following illustration.

This definition of location semantics, represented by PoI distributions, is applicable to all cases, including regions characterized by sparse PoIs. The semantics of a location without any PoI nearby will be a zero vector. Nevertheless, the zero vector semantics of these areas remains as meaningful as non-zero ones, as it characterizes the uninhabited character of these locations.

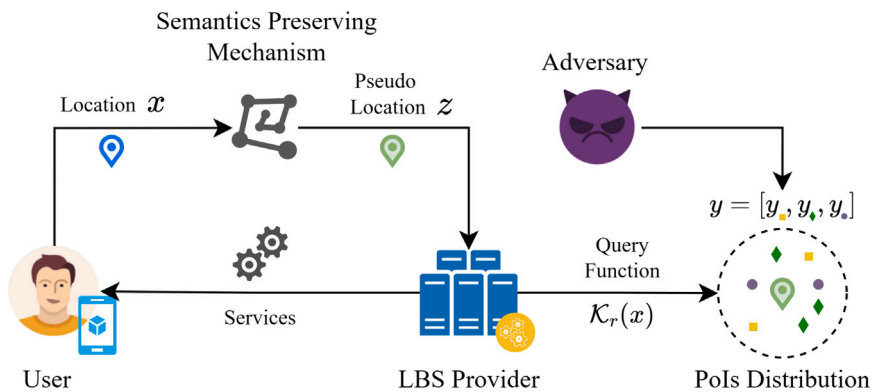

Fig. 2. System overview of LBS with adversaries and semi-honest SP involving in.

4.2. Threat model

As shown in Fig. 2, we assume that the adversary can access the pseudo-location z from the LBSP's database or public transmission channel, while not interfering with the normal interaction between users and LBSP. Besides, the adversary fully grasps the location obfuscation mechanism performed on the client side, as well as prior knowledge about the probability distribution of users' actual location x, denoted as $\pi(x)$. How the prior knowledge is derived by the adversary is out of scope in this paper.

4.3. Privacy objective

As illustrated in the introduction section, the semantics which is represented by the distribution of nearby PoIs can reflect the attributes of locations users visit, indicating the characteristics about users' working, visiting or other activities. Therefore, it is necessary to reduce the semantic information that could be leaked from the user's location. A straightforward approach is to obfuscate the true location to a pseudo-location, thus exhibiting semantics different from the true location's semantics. For each location, a random location obfuscation mechanism can derive a probability distribution of the obfuscated semantics. If the probability distributions of obfuscated semantics across different locations exhibit significant variations, these distributions can likewise reflect the characteristics of the locations. Therefore, the goal is not only to conceal the true semantics of locations but also to obscure the probability distributions of their obfuscated semantics into the crowd. We propose to protect location semantics privacy under the paradigm of differential privacy, where location obfuscation is rigorously specified by probability constraints, so that the true semantics is hidden behind randomness, escaping from accurate observations by adversaries and LBSPs.

Semantic Indistinguishability. We import the notion of differential privacy into the semantic domain to define the location semantic privacy guarantee formally. Under the framework of LPPM, the obfuscation of location x should follow the differential privacy guarantee through the well-crafted location obfuscation mechanism $\mathcal{M}$. Specifically, any two locations x_1, x_2 should be obfuscated with similar probabilities to locations whose semantics are the same.

Definition 3 (*ϵ-Semantic-Indistinguishability*)**.** Let $\mathcal{Y}_i, 1 \le i \le N$, be the set of values for the i-th category. Given the semantic query function $\mathcal{K}_i : \mathcal{X} \to \mathcal{Y}_i$, a probabilistic location obfuscation mechanism $\mathcal{M} : \mathcal{X} \to \mathcal{X}$ provides ϵ-Semantic Indistinguishability (ϵ-Sem-Ind) if $\forall y_i \in \mathcal{Y}_i$, $\forall x_1, x_2 \in \mathcal{X}$, the following condition is satisfied:

$$\frac{\Pr[\mathcal{K}_i(\mathcal{M}(x_1)) = y_i]}{\Pr[\mathcal{K}_i(\mathcal{M}(x_2)) = y_i]} \le e^{\epsilon}. \tag{7}$$

This definition requires that the probability distribution indistinguishability for semantic values should hold for each semantics dimension. To simplify the expression of multi-dimensional location semantics in subsequent formulas, subscript notation for the category index is omitted in the following. Apparently, whether Sem-Ind holds or not is determined by both the obfuscation mechanism $\mathcal{M}$ and the semantic query function $\mathcal{K}$. This is what differs most from Geo-Ind, which imposes direct constraints solely on $\mathcal{M}$.

The absence of prior distribution $\pi(x)$ in Eq. (7), which is similar to Geo-Ind and other LDP guarantees, shows that the privacy preserving provided by Sem-Ind should hold regardless of any prior distribution. The goal of Sem-Ind is to make it difficult for an attacker to make further enhancements to the prior distribution after observing the location semantics y derived from the pseudo-location z, rather than completely preventing the attacker from inferring the user's true locations based on the semantics. This constraint on information leakage originates from the inherent randomness introduced by the obfuscation mechanism.

Theorem 2. *If the obfuscation mechanism $\mathcal{M}$ satisfies ϵ-Sem-Ind, then the information gain of Bayesian adversaries from the posterior distribution to the prior knowledge of location x is bounded.*

Proof. The proof is similar to the Equation (19) in Yu et al. (2017). Given any prior knowledge $\pi(x)$, the posterior distribution of location x given that the semantic observation O_s equals y should satisfy

$$\Pr[X = x | O_s = y] \le e^{\epsilon}\pi(x), \tag{8}$$

which implies that the ratio between the posterior distribution $\Pr[x|y]$ and the prior distribution $\pi(x)$ is bounded within e^{ϵ}. Thus the proof is completed. □

4.4. Problem objective

Utility Metric. Besides the privacy guarantee, the utility (aka service quality) should also be considered. The loss of utility can be regarded as an obfuscation error, which is the function of the distance between the true and the pseudo location (Bordenabe et al., 2014). So the average impact of location obfuscation mechanisms on utility can be evaluated by the expected error. Thus we have

$$\begin{aligned} ExpUL &= \mathbb{E}_{\pi,\mathcal{M}}[d(x, \mathcal{M}(x))] \\ &= \sum_{x,z\in\mathcal{X}} \pi(x) f(z|x) d(x, z), \end{aligned} \tag{9}$$

where d denotes the euclidean distance between two locations and f denotes the probability specified by the obfuscation mechanism $\mathcal{M}$.

Privacy Metric. While the privacy parameter ϵ specifies an upper bound of multiplicative distance, it is crucial to empirically evaluate the performance of privacy preservation using other metrics. We use Jensen–Shannon divergence which is founded on the well-known KL

divergence (Tu et al., 2017; Li and Li, 2009) to evaluate the similarity between probability distributions of semantics. The KL Divergence of two semantic obfuscation probability distributions P and Q is defined as

$$\mathbb{D}_{KL}(P \parallel Q) = \sum_{y \in \mathcal{Y}} P(y) \cdot \ln(\frac{P(y)}{Q(y)}), \tag{10}$$

while JS Divergence can be obtained as

$$\mathbb{D}_{JS}(P \parallel Q) = \frac{1}{2}\mathbb{D}_{KL}(P \parallel Q) + \frac{1}{2}\mathbb{D}_{KL}(Q \parallel P). \tag{11}$$

Intuitively, the closer the distance between probability distributions, the harder it is for an attacker to distinguish different locations by their obfuscation probabilities. As well as the divergence, the information gain brought by observation after obfuscation should be measured in the adversary's view. Here we introduce the error of an optimal inference attack carried out by the Bayesian adversary, which aims to minimize the expected distortion between the users' actual locations and the inferred locations over the probability of posterior distribution. The inference error can be computed as follows:

$$\begin{aligned} AdvErr &= \min_{\hat{x}} \sum_{x \in \mathcal{X}} \Pr[x|y] d(x, \hat{x}) \\ &= \min_{\hat{x}} \sum_{x \in \mathcal{X}} \frac{\pi(x) \Pr[y|x]}{\sum_{z \in \mathcal{X}} \pi(z) \Pr[y|z]} d(x, \hat{x}), \end{aligned} \tag{12}$$

where $\Pr[y|x]$ can be computed as $\sum_{z \in \Phi(y)} f(z|x)$, and $\Phi(y)$ denotes the location set whose semantic value is y.

Theorem 3. *When Sem-Ind is satisfied, there is a lower bound for inference error, which is* $e^{-\epsilon} \min_{\hat{x}} \sum_x d(\hat{x}, x)$.

The proof is similar to Equation (25) in Yu et al. (2017).

5. Semantic-indistinguishability location obfuscation mechanism variants

In this section, we will present several location obfuscation mechanisms. Firstly, a straightforward adaptation of the Geo-Ind-based mechanism will be introduced, revealing the relationship between Sem-Ind and Geo-Ind. Then, as a baseline, we formulate a linear programming problem targeting the optimal utility objective under the privacy guarantee. Besides, greedy heuristic approaches are proposed for further efficiency improvement.

5.1. Adaptation of geo-indistinguishability

It is quite trivial to prove that the constraints of Geo-Ind on location obfuscation can be adapted to that of Sem-Ind.

Theorem 4. *If an obfuscation mechanism $\mathcal{M}$ satisfies ϵ-Geo-Ind, and D is the maximum distance between any two locations (diameter of the region), then $\mathcal{M}$ satisfies $(\epsilon \cdot D)$-Sem-Ind.*

Proof. According to the definition of ϵ-Geo-Ind, given any location x_1, x_2, the distance of obfuscation probabilities is bounded by $\epsilon d(x_1, x_2)$, which is always smaller than $\epsilon \cdot D$. Thus, ϵD-differential privacy is satisfied. Subsequently, based on the Post-Processing properties (Dwork and Roth, 2014) of differential privacy, the application of the semantic query function $\mathcal{K}$ after $\mathcal{M}$ will not violate the DP properties. Note that the Post-Processing property holds no matter whether y is 1-dimensional or multi-dimensional. Thus the proof is completed. □

Due to the strict privacy guarantee provided by Geo-Ind, we can leverage several existing location obfuscation mechanisms derived from Geo-Ind originally including Planar Laplacian (PL) (Andrés et al., 2013), to construct Sem-Ind mechanisms with small adaption of scaling the privacy parameter ϵ. However, the guarantee of Geo-Ind is stricter than the requirement of Sem-Ind. In Sem-Ind, what we care about is the overall probabilities of obfuscation to locations with the same semantic value, rather than the same location. As it is highly possible for more than one location in the region to share the same semantic value, there should be a difference between obfuscation probabilities to different locations with an identical semantic value.

For example, consider that there are only two locations in $\mathcal{X}$, namely x_1, x_2. The counts of restaurants nearby are 1, 1, respectively, namely $\mathcal{K}(x_1) = 1, \mathcal{K}(x_2) = 1$. Suppose a 0-Geo-Ind mechanism requires that given any pseudo-location z, there must be $f(z|x_1) = f(z|x_2) = c$, where c is a constant. So the probabilities of obfuscation from any location x to locations with semantic value 1, that can be denoted as $\Pr[\mathcal{K}(\mathcal{M}(x)) = 1] = f(x_1|x) + f(x_2|x)$, are the same, which means $\Pr[\mathcal{K}(\mathcal{M}(x_1)) = 1] = \Pr[\mathcal{K}(\mathcal{M}(x_2)) = 1]$. Thus 0-Sem-Ind is also satisfied. However, 0-Sem-Ind does not necessarily require 0-Geo-Ind, but only $f(x_1|x_1)+f(x_2|x_1) = f(x_1|x_2)+f(x_2|x_2)$. Thus, the case when $f(x_1|x_1) \neq f(x_1|x_2)$ could be raised, such as $f(x_1|x_1) = f(x_2|x_2) = 1/3$ and $f(x_1|x_2) = f(x_2|x_1) = 2/3$. This condition breaches 0-Geo-Ind, but holds 0-Sem-Ind. Geo-Ind is sufficient for Sem-Ind, but is not necessary for Sem-Ind.

In the subsequent sections, we will present several mechanisms designed to exploit more utility than Geo-Ind mechanisms while still ensuring Sem-Ind.

5.2. Optimal semantic-indistinguishability

By combining the privacy and utility objectives together, we can formulate a linear optimization problem for Sem-Ind (OPT-SEM-IND) to obtain f, which takes the expected utility loss as the minimization objective. The goal is to find an obfuscation probability distribution f to minimize the expected utility loss $ExpUL$ under the prior distribution $\pi(x)$ of the specific user while not violating the constraints of Sem-Ind.

$$(\mathbf{OPT-SEM-IND}) \quad \arg\min_{\mathbf{f}} \sum_{x,z \in \mathcal{X}} \pi(x) f(z|x) d(x, z) \tag{13}$$

$$\text{s.t.} \quad \frac{\sum_{z \in \Phi(y)} f(z|x_1)}{\sum_{z \in \Phi(y)} f(z|x_2)} \leq e^{\epsilon}, \ \forall\ x_1, x_2 \in \mathcal{X}, y \in \mathcal{Y} \tag{14}$$

$$\sum_{z \in \mathcal{X}} f(z|x) = 1, \ \forall\ x \in \mathcal{X} \tag{15}$$

$$f(z|x) \geq 0, \ \forall x, z \in \mathcal{X} \tag{16}$$

As is mentioned in the definition of Sem-Ind, the DP constraints (14) should cover each dimension y_i of the semantics, which is omitted in the formulation of the linear programming problem for clarity. The OPT-SEM-IND is defined to search in the solution space beyond that of Geo-Ind, trading privacy defined in Geo-Ind with utility gain. The solution of this linear programming problem can be found by some commonly-used optimization solvers like Gurobi.

Approximation. The computational complexity of solving OPT-SEM-IND in practice is dominated by the count of constraints, which is $O(|\mathcal{X}|^2|\mathcal{Y}|)$. Specifically, there exists a single constraint for each tuple (x, z, y) comprising two locations and one semantic value, resulting in a quadratic growth in the problem scale concerning the number of locations. Furthermore, when N-dimensional semantics y are handled, the number of constraints is indeed $O(N|\mathcal{X}|^2|\mathcal{Y}|)$.

To scale down the number of constraints, the pairs of locations to be compared in OPT-SEM-IND should be restricted. We formulate all the comparison relationships as a graph $\mathcal{G}(\mathcal{X}, \mathcal{E})$, whose vertices represents location x and each edge $(x_1, x_2) \in \mathcal{E}$ corresponds to the required comparing operation in OPT-SEM-IND between the locations it connects. Then diameter-2-critical graph (Caccetta and Häggkvist, 1979; Wang et al., 2016) is introduced to approximate the optimal solution under the constraints of comparison between selected pairs. The comparison graph is constructed under the condition that the maximum distance (diameter) between any two vertices is less than

Algorithm 1 Nearest Semantic Sampling NSS

Input: $\mathcal{X}, \mathcal{Y}, x, \mathcal{K}, \rho$;
Output: x_i;
Sample new semantic value y from $\mathcal{Y}$ by distribution ρ;
for $x_i \leftarrow BFS(x)$ **do**
  if x_i is the nearest location with semantic $\mathcal{K}(x_i) = y$ **then**
    return x_i
  end if
end for

or equal to 2, and any deletion of the edge will increase the diameter. In practice, we choose the location 0 as the central vertex of the graph, connecting any other location with it.

Theorem 5. *If the obfuscation distribution f derived from the comparison specified by graph $\mathcal{G}(\mathcal{X}, \mathcal{E})$ which is a diameter-2-critical graph, satisfies*

$$\sum_{z\in\Phi(y)} f(z|x_1) \le e^{\epsilon} \sum_{z\in\Phi(y)} f(z|x_2), \forall\ (x_1, x_2) \in \mathcal{E},\ \ y \in \mathcal{Y} \tag{17}$$

then, f satisfies 2ϵ-Sem-Ind.

The theorem can be proved following Theorem 1 in Wang et al. (2020) and is quite trivial. Leveraging the properties of diameter-2-critical graphs, the number of comparisons can be reduced to $O(|\mathcal{E} \parallel \mathcal{Y}|) = O(|\mathcal{X} \parallel \mathcal{Y}|)$ in 1-dimensional semantics and $O(N|\mathcal{X} \parallel \mathcal{Y}|)$ in the case of N-dimensional semantics.

5.3. Heuristic mechanism

Even if the number of constraints is reduced to a linear level, solving the approximated OPT-SEM-IND is feasible only when $|\mathcal{X}|$ is small, typically limited to a few hundred locations. In this section, two heuristic mechanisms will be given to provide a Sem-Ind guarantee by conducting a straightforward search and sampling. These approaches are considerably more efficient than linear programming methods. We will first demonstrate these two mechanisms, **NSS** and **ESSA**, for the case of 1-dimensional semantic value and then integrate them into the mechanism **ISS** for the multi-dimensional case.

Nearest Semantic Sampling(NSS). As is given in Alg. 1, the intuition of NSS is that to restrict the multiplicative distance between probabilities of sampling the same semantics, we can sample semantic values by an identical distribution. One feasible semantic sampling approach is uniform sampling. To minimize the distance between pseudo-locations and raw locations, NSS samples the pseudo semantics y according to the probability distribution of semantics in the whole region, denoted as ρ. Intuitively, for any given location, it has a higher probability of being obfuscated to locations with more common semantics, instead of ones with more rarely seen semantics, thus geographically closer locations are preferred under the distribution ρ. After the pseudo semantics y is defined, a Breadth-First-Search(BFS) is launched from the actual location x until the first pseudo-location z is found with semantic y. NSS is quite straightforward to provide obfuscation considering the minimization of utility loss under the Sem-Ind guarantee in a greedy manner. In fact, NSS satisfies 0-Sem-Ind, since for all the x, $\Pr[\mathcal{K}(\mathcal{M}(x)) = y] = \rho(y)$.

Exponential Semantic Sampling Adjustments(ESSA). ESSA takes the influence on utility into account when choosing semantics. The intuition is to first adopt the exponential mechanism to generate a probability distribution providing location differential privacy, which is $f(z|x_1) \le e^{\epsilon} f(z|x_2)$. Based on the Post-Processing properties, this distribution also satisfies Sem-Ind. Then, for each x, we reallocate the probabilities $f(z|x)$ of all the $z \in \Phi(y)$ given y in terms of the corresponding obfuscation error. Greedily, the obfuscation bringing the least error $d(z,\ x)$ will receive all the probabilities of obfuscation from x to $\Phi(y)$.

Algorithm 2 Exponential Semantic Sampling Adjustments ESSA

Input: $\mathcal{X}, \mathcal{Y}, x, \mathcal{K}, \epsilon$;
Output: z;
$p \leftarrow empty\ array$;
for $x_i \leftarrow \mathcal{X}$ **do**
  $p[x_i] \leftarrow \exp(\frac{-\epsilon d(x,x_i)}{2D(\mathcal{X})})$;
end for
Normalize p;
for $y \leftarrow \mathcal{Y}$ **do**
  $\Phi(y) \leftarrow\ all\ the\ x_i\ with\ \mathcal{K}(x_i) = y$;
  $p_y \leftarrow \sum p[\Phi(y)]$
  $x_{min} \leftarrow arg\,min_{x_i}\ d(x_i, x)$;
  $p[\Phi(y)] \leftarrow 0$;
  $p[x_{min}] \leftarrow p_y$;
end for
Sample z from $\mathcal{X}$ following the distribution p;
return z

Algorithm 3 Iterative Semantic Sampling ISS

Input: $\mathcal{X}, \mathcal{Y}, x, \mathcal{K}, \mathcal{M}, N$;
Output: z;
$z \leftarrow x$;
for $i \leftarrow N$ **do**
  $z \leftarrow \mathcal{M}(\mathcal{X}, \mathcal{Y}, \mathcal{K}, z)$ on dimension i;
end for
return z

In the exponential mechanism, each location is sampled from $\mathcal{X}$ with the following distribution:

$$f(z|x) = w_x e^{\frac{-\epsilon d(x,z)}{2\Delta u}} \tag{18}$$

$$w_x = 1/\sum_{z\in\mathcal{X}} e^{\frac{-\epsilon d(x,z)}{2\Delta u}}, \tag{19}$$

where $d(x, z)$ is the distance, and Δu is the sensitivity of utility function. Here, Δu is the diameter of $\mathcal{X}$, denoted as $D(\mathcal{X})$. The exponential mechanism yields high-quality utility assurances since pseudo-locations are discounted exponentially as their distance to actual location increases.

Theorem 6. *ESSA satisfies ϵ-Sem-Ind.*

Proof. It has been proved that exponential mechanism which applies obfuscation distance $d(x, z)$ as utility function and diameter as sensitivity satisfies ϵ-differential privacy (Yu et al., 2017). Then the processing of query function $\mathcal{K} : \mathcal{X} \to \mathcal{Y}$ after $\mathcal{M}$ will not violate ϵ-Sem-Ind, which means

$$\frac{\sum_{z\in\Phi(y)} f(z|x_1)}{\sum_{z\in\Phi(y)} f(z|x_2)} \le e^{\epsilon} \tag{20}$$

Note that the reallocation will not mute the value of $\sum_{z\in\Phi(y)} f(z|x)$. Thus ϵ-Sem-Ind is also satisfied for ESSA. The proof is completed. □

It is straightforward to figure out that the time complexities of both NSS and ESSA are $O(|\mathcal{X}|)$.

Both of the above heuristic mechanisms provide Sem-Ind only for a one-dimensional semantic vector. When working with N-dimensional semantic vectors, generating a pseudo-location z that accounts for only one dimension can cause disturbances in other dimensions of the semantic vectors. However, these disturbances might not strictly

Table 1
Parameters setup in preprocessing datasets.

Parameters	Settings		
Dataset	GeoLife		
City(District)	Beijing(Haidian)		
Region Northwest	(39.9870,116.2952)		
Region Southeast	(39.9450,116.3539)		
Location size	200 m × 200 m		
Query range for nearby PoIs	200 m		
PoI density level	Low	Middle	High
PoI category	Scenic spot	Public facility	Restaurant
PoI count	41	453	1550

conform to the Sem-Ind principle. To address this issue, we propose the naive Iterative Semantic Sampling(ISS), as is given in Alg. 3.

Iterative Semantic Sampling(ISS). To guarantee Sem-Ind for a multidimensional semantic vector, each dimension is perturbed by location obfuscation successively and separately. Specifically, for all N semantic categories, each dimension i is perturbed by one-dimensional obfuscation mechanisms $\mathcal{M}$ (either NSS or ESSA) in iterations. In each iteration for dimension i, the obfuscation is conducted on the pseudo-location z produced by the last iteration for dimension $i-1$. As a whole, the ISS samples through a probabilistic location transition chain to guarantee the multi-dimensional semantic privacy of the obfuscation procedure. It is trivial to prove that ISS satisfies ϵ-Sem-Ind for dimensions N using Post Process properties of differential privacy.

6. Experimental evaluation

In this section, we comprehensively analyze Sem-Ind on a real-world datasets and evaluate the effectiveness of various proposed algorithms in terms of utility and privacy.

6.1. Experimental setup

We adopt the GeoLife dataset. It contains trajectory data collected from 178 users over a period of more than four years (from April 2007 to October 2011) and contains 17,621 trajectories. Most of the users move mainly in north-west Beijing, performing routine tasks. These trajectories were recorded by a variety of GPS loggers and GPS phones, with different sampling rates, 91% of which were sampled every 1 to 5 s or every 5 to 10 m per point (Zheng et al., 2011).

First, we select the area for subsequent experiments based on the latitude and longitude ranges provided in Table 1, which is located in the Haidian District of Beijing. Then, we uniformly divide the rectangle area into grids, with each grid representing a 200 m x 200 m location area. The entire area is divided into 500 locations.

We filter the user records in the dataset according to the geographical area outlined in Table 1. To avoid uneven sampling frequency of the user locations in the original data, we resample each user's trajectory data with a uniform sampling interval of 2 min. Then we filter out those users who contributed fewer than 50 location records, who can be seen as less informative data contributors than others.

In addition to user location data, we use a private list of PoIs within the area in Beijing, which contains approximately 350,000 PoI records. Each record contains information of one PoI, including geographic coordinate, base-type, sub-type and concrete type. In the following evaluation, we adopt the base-type as the PoI category, which contains 19 distinct possible values. To assess the impact of varying distribution characteristics of different PoI categories on location semantic privacy protection, we select three PoI categories with different distribution densities. The categories and quantities are presented in Table 1.

All experiments in this paper are written in Python, running on a 16-core Intel E5-2620 v4 @ 2.10 GHz with 64G of RAM. Common third-party libraries such as pandas and folium are used in the coding.

6.2. Evaluate city semantic distribution

To give an overview of the semantic distributions, the three selected PoI categories are visualized on the map in Fig. 3(a) to illustrate their spatial distribution patterns, and the distribution of location counts characterized by the semantics of these three categories are illustrated separately in Fig. 3(b)(c)(d). The PoIs with low, middle and high density are visualized using purple, orange, and blue markers respectively.

As is shown in Fig. 3(a), restaurants, denoted by blue markers, represent the highest quantity and density in the region. Their spatial distribution exhibits non-uniform patterns, with significant clustering along road networks and within commercial zones, leading to high density in localized areas. In particular, with location grids plotted at 200-m resolution on the map, it can be observed that certain locations lack any restaurants in their proximity. Compared to restaurants, the other two PoI categories show sparser distributions, especially scenic spots, with 41 spots mainly concentrate in approximately 10 local regions.

As shown in Fig. 3(b)(c)(d), 95% of locations have no more than 2 scenic spots, 7 public facilities, and 27 restaurants in their vicinity. For each of the PoI categories, there exists more than one location whose count of the nearby PoIs, aka location semantics, is unique among all locations. These semantics reflect location attributes and could result in downgraded privacy level, with locations having rare semantics posing potentially higher privacy risks.

The basic view of Sem-Ind as defined in Section 4.3 is to treat semantically identical locations as an equivalent set, and to require indistinguishability of probabilities obfuscating any location to the same semantic set. Thus, the true location semantics could be hidden.

6.3. Evaluate privacy and utility

The privacy and utility performances of all mechanisms are evaluated empirically based on user location records from the GeoLife datasets. The evaluated privacy preserving mechanisms are as follows:

- Planar Laplacian (PL) (Andrés et al., 2013).
- Optimal Location Differential Privacy (OPT-LOC-DP): Obtain obfuscation distribution f by solving a linear optimization problem, where the goal is to minimize the expected utility loss under the constraints of location differential privacy. Note that OPT-LOC-DP is slightly different from optimal geo-indistinguishability (Bordenabe et al., 2014).
- Optimal Semantic-Indistinguishability (OPT-SEM-IND): Similar to OPT-GEO-IND, but the constraints it follows is Sem-Ind instead.
- Nearest Semantic Sampling (NSS): Sample each semantic uniformly and search the nearest satisfying location. In multi-dimensional cases it is embedded in ISS, namely NSS+ISS.
- Exponential Semantic Sampling Adjustment (ESSA): Apply the exponential mechanism over the whole region first, then adjust the probabilities within each semantic equivalent set. In multi-dimensional cases it is embedded in ISS, namely ESSA+ISS.

In the evaluation, we adopt one-dimension vector as the location semantics, which is the count of nearby PoIs with the category introduced in Table 1. The utility loss is measured by expected distance and the privacy gain is measured by JS divergence and inference error, as demonstrated in Section 4.4. It should be emphasized that all three metrics are obtained through empirical evaluation, meaning the numerical results are calculated from sampled data based on probability distributions, rather than directly derived from the probability distributions themselves. In the experiment, for each location, we sample 100 pseudo-locations to compute the mean utility loss, JS divergence and inference error.

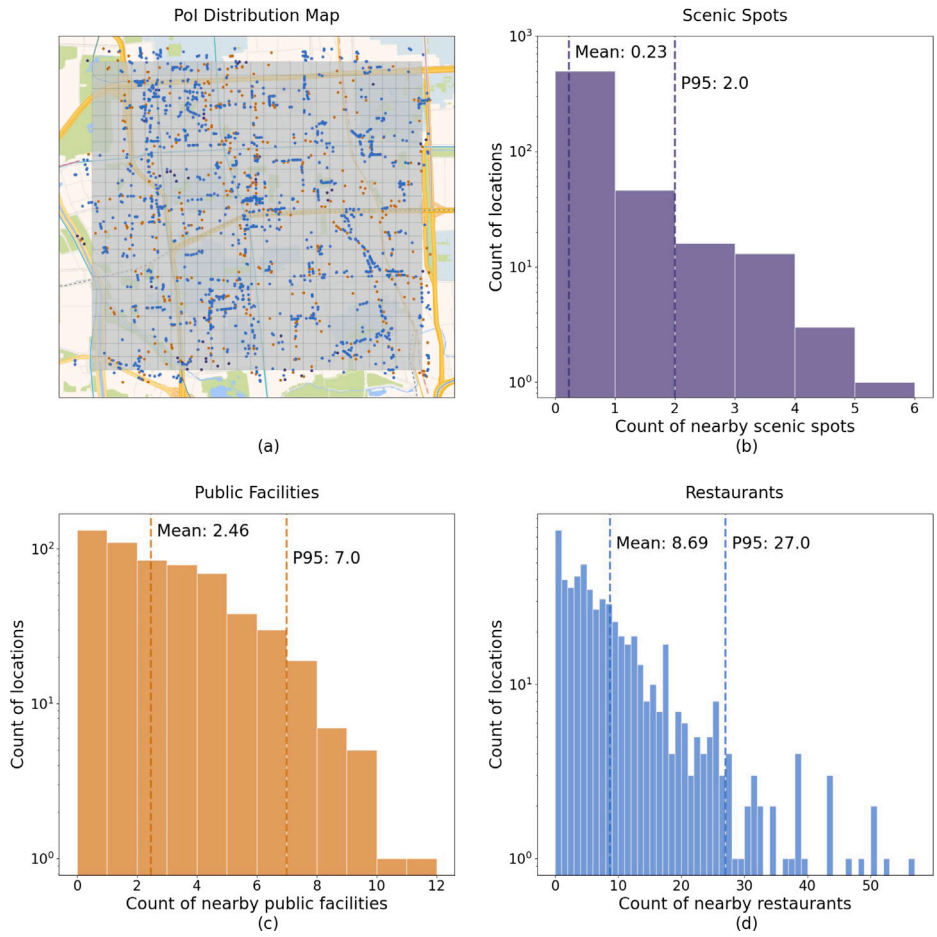

Fig. 3. The spatial distribution and statistics of semantics in the region in Haidian District.

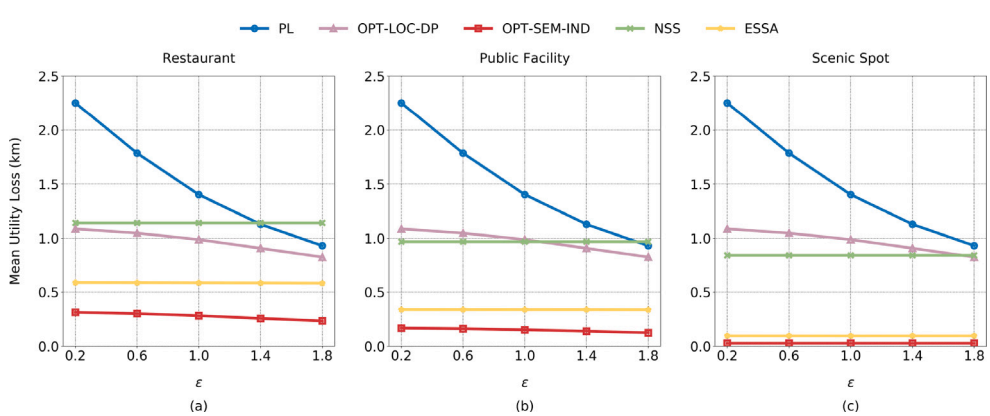

Fig. 4. Variation of mean utility loss against parameters ϵ.

Fig. 4 illustrates the average utility loss of the five algorithms. Given that PL and OPT-LOC-DP are designed and implemented without considering location semantics, they demonstrate consistent performance. PL yields the largest obfuscation errors, exceeding 0.8 km, which is significantly higher than the other mechanisms evaluated when ϵ is less than 1.4. When ϵ exceeds 1.4, PL exhibits lower utility loss than NSS. The OPT-SEM-IND mechanism exhibits the most favorable utility performance, as Sem-Ind is a more relaxed privacy guarantee than location DP, resulting in a considerable utility advance. Besides, comparing the utility loss among three PoI categories, it can be seen that lower density and count of PoIs corresponds to reduced utility loss. The optimal utility loss is almost zero when semantics about scenic spot is protected using OPT-SEM-IND, as is shown in Fig. 4(c). This occurs because locations with semantics 0 exist in the vicinity of almost every location. As Sem-Ind blurs the difference of the probability by which any two locations are obfuscated to one semantic equivalent set, OPT-SEM-IND tends to obfuscate location to a nearby one with similar probabilities. As a result, locations of any semantic have a substantial probability of being obfuscated to the widespread 0-semantic locations, which hides their actual semantics. The utility increase brought by OPT-SEM-IND alongside the increase of ϵ is smaller than that of OPT-LOC-DP since more privacy can be preserved by OPT-LOC-DP. The NSS can be

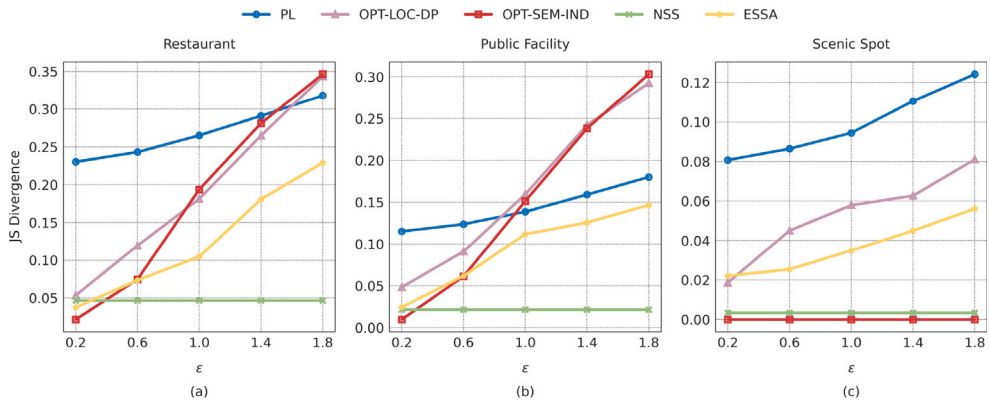

Fig. 5. Variation of JS divergence against parameters ϵ.

regarded as a Sem-Ind with $\epsilon = 0$ constantly, showing a lower level of error than PL. Our proposed exponential mechanism-based ESSA has a utility loss between two linear programming-based algorithms, remaining small variations in the cases of all three PoI categories. In Fig. 4(a), the utility loss of ESSA decreases from 0.587 to 0.582. Besides, ESSA and NSS also have the same variation phenomenon with different PoI densities.

The performance of these mechanisms in terms of privacy protection is illustrated in Figs. 5 and 6 based on JS divergence and inference error respectively. Intuitively, lower divergence and higher inference error indicate superior privacy preservation.

Fig. 5 illustrates the variation of JS divergence between the semantic distributions obtained after obfuscation for each location concerning the privacy parameter ϵ. It can be observed from Fig. 5(a)(b)(c) that the divergence decreases as PoI distribution becomes sparser, corresponding to a reduced set of possible location semantic values. Considering both the value and the ratio of divergence variation, OPT-LOC-DP and OPT-SEM-IND appear almost the same when the distribution of restaurants and public facilities are viewed as semantics. The JS divergence of both OPT-LOC-DP and OPT-SEM-IND are significantly affected by the variation of ϵ. This outcome can be attributed to the fact that both mechanisms prioritize minimizing utility loss at the expense of privacy to the greatest extent. An exception occurs when considering only scenic spots as PoIs. Due to the extreme sparsity of scenic spots in the region, the obfuscation probability f obtained by OPT-SEM-IND provides deterministic outputs for each input location. Specifically, any location x would be obfuscated to a pseudo-location deterministically whose semantic value is either 0 or 1. Note that when ϵ is set in our evaluation, the PL satisfies $\epsilon \cdot D$-Sem-Ind, as detailed in Section 5.1. The gap between PL and the other two optimization mechanisms highlights the potential of PL to prioritize privacy over utility. Regarding NSS and ESSA, both mechanisms exhibit comparable performance concerning JS divergence. While intuition suggests that NSS should yield the same probability distributions across all locations, the calculated JS divergence is non-zero. We attribute this discrepancy to insufficient sampling before the divergence computation phase of our experiments.

The variation of another privacy metric, inference error after obfuscation, is depicted in Fig. 6. There is a downward trend of inference error as ϵ increases for all the mechanisms except NSS and OPT-SEM-IND with scenic spots, among which OPT-LOC-DP and PL exhibit greater exposure of private information when subjected to inference attacks. In Fig. 6(a), when ϵ is 1.8, the inference error after applying PL is only about 0.5, which means the inferred location is 2–3 locations away from the actual one. When restaurant or public facility is chosen as PoI, OPT-SEM-IND brings about larger errors than two heuristics mechanisms when applying the same privacy parameter ϵ. Similar to our findings in the JS divergence evaluation, OPT-SEM-IND shows constant performance for scenic spots due to the same underlying cause. Among all algorithms except NSS, ESSA shows the slowest decrease in inference error as ϵ increases.

In order to examine the performance of these mechanisms in terms of utility in the context of multidimensional semantics, we evaluate five dimensions, from 1 to 5, respectively. The five semantic dimensions include the three PoI categories mentioned in the experimental setup, plus two additional categories: accommodation and healthcare. In this evaluation, ϵ is set to be 0.7. The obtained result is shown in Fig. 7. Obviously, PL and OPT-LOC-DP are not affected by the number of dimensions. As dimensions increase, the privacy requirement for Sem-Ind enhances, so OPT-SEM-IND, NSS, and ESSA all cost more utility to realize the privacy guarantee. Among them, NSS exhibits the most rapid decrease in utility as the semantic dimension expands, while ESSA and OPT-SEM-IND show similar rates of decrease, which are slower than NSS. Moreover, OPT-SEM-IND even brings smaller utility loss with the highest dimension compared with other methods which are conducted in low-dimension cases. It incorporates all dimensions into the constraints defined by Sem-Ind to find the optimal solution in the vast solution space.

6.4. Evaluate mechanism efficiency

To evaluate the running efficiency, we design and conduct experiments to empirically measure the running time of these algorithms. We take the dominant factors including location size $|\mathcal{X}|$ and dimensions N into account when measuring running time. It should be noted that when measuring the running time of OPT-LOC-DP and OPT-SEM-IND, we first launch the optimization solving process once and then sample the probability distribution f 1000 times. PL, NSS and ESSA are also run 1000 times, to ensure fair comparison with optimization-based mechanisms. Besides, in our implementation of OPT-LOC-DP, we utilize the diameter-2 critical graph to reduce the time complexity to the quadratic level of the number of locations. In the experiments shown in Fig. 8(a), we examine the running time of all five algorithms concerning the variation of $|\mathcal{X}|$. From the northwest corner indicated in 1, $|\mathcal{X}|$ locations are selected in a configuration that approximates a square shape. The count of nearby restaurants is chosen as the one-dimension location semantics. As is shown in the figure, the running time of OPT-LOC-DP increases at a higher rate than that of OPT-SEM-IND. When the count of locations is up to 500, it takes OPT-LOC-DP about 21 s. As analyzed in Section 5.2, the scale of magnitude of the OPT-SEM-IND constraints is proportional to the number of locations in the region, and the factor is the number of distinct semantic values, which is generally significantly smaller than the number of locations. In comparison with the aforementioned two algorithms, the computation time required by PL, USS, and ESSA is negligible. This results indicate that PL, USS, and ESSA algorithms can be effectively implemented on users' mobile devices in practical settings without substantial degradation of service experience.

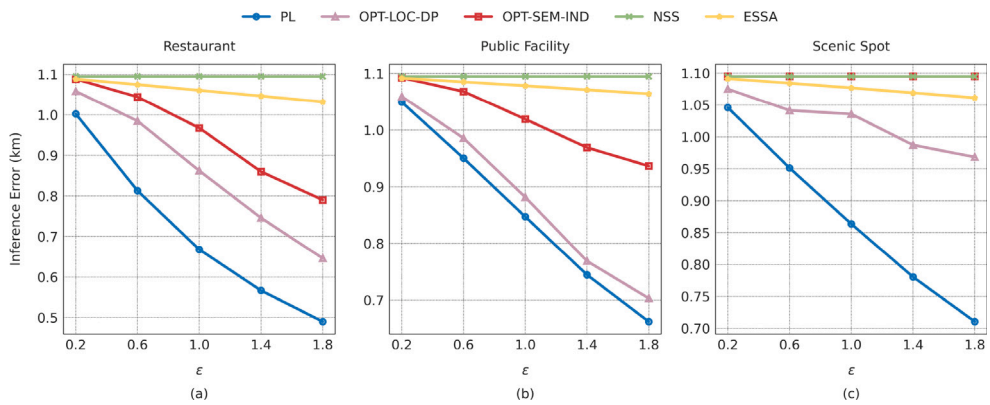

Fig. 6. Variation of inference error against parameters ϵ.

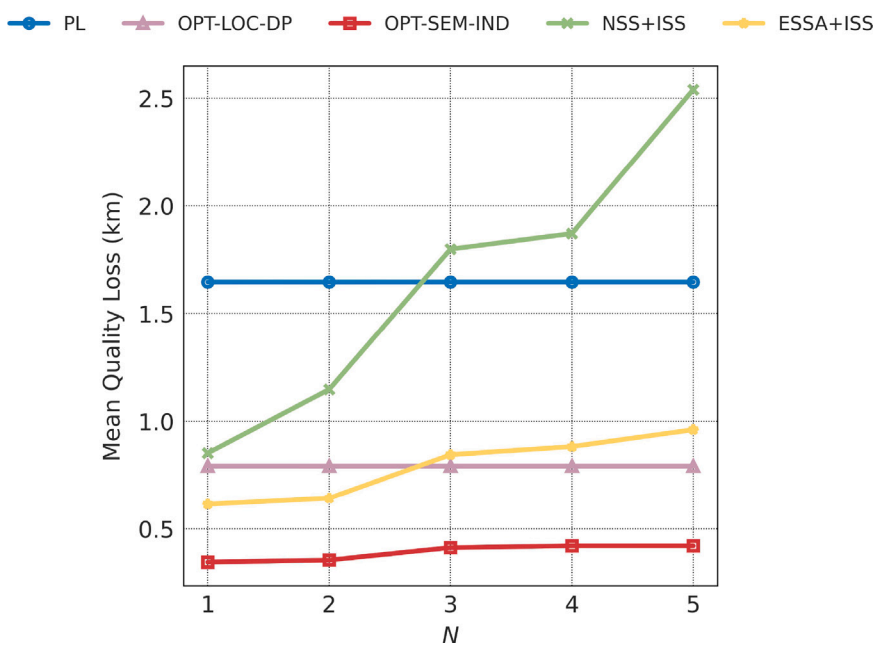

Fig. 7. Variation of mean utility loss against dimensions.

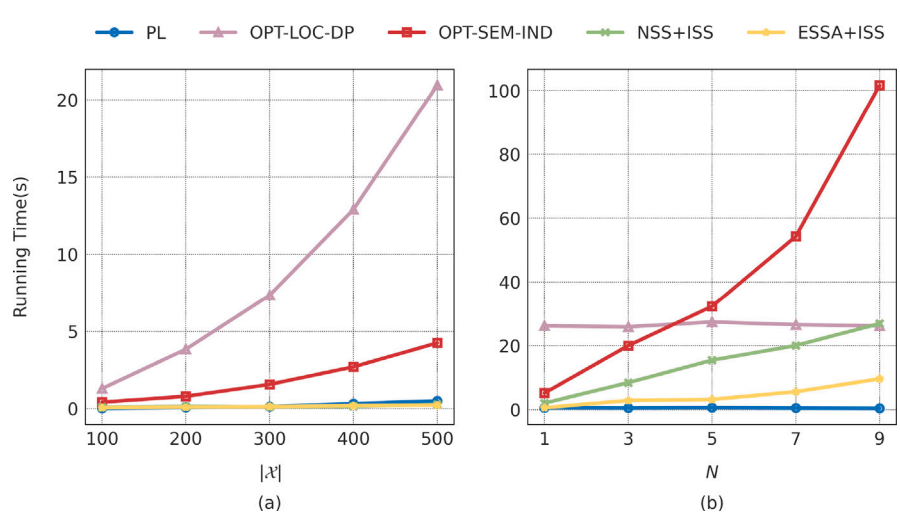

Fig. 8. The running time for different $\mathcal{X}$ and dimensions.

Fig. 8(b) illustrates the variation of running time when semantic dimension increases. We set the $|\mathcal{X}|$ to 500, incorporate the five PoI categories used to obtain Fig. 7, as well as 4 PoI categories: Shopping, Living, Education and Company. The running time of PL and OPT-LOC-DP remains unchanged, as they are independent of dimensions. When comparing NSS and ESSA, both of them are $O(|\mathcal{X}|)$. The running time of ESSA also increases with the dimension of semantic data, but at a slower rate than NSS. OPT-SEM-IND exhibits the longest execution time in the figure, requiring 100 s when the semantic dimension is 9. The majority of the time required is consumed by solving the optimization problem. For real-world deployment of OPT-SEM-IND, the optimization problem can be solved on high-performance servers, and more sampling times can be performed after obtaining f to amortize the computational overhead. In general, heuristic mechanisms have the potential to discover solutions comparable to that of optimization-based mechanisms, while requiring less computing time and resources.

7. Practicability analysis

This paper proposes Sem-Ind and three location semantics privacy preserving mechanisms. In practical scenarios, these mechanisms operate on the user side, predominantly mobile devices. With privacy protection mechanisms enabled, users' actual locations are first obfuscated by these mechanisms. Then the pseudo-location will be sent to the LBSP. Protecting location semantics privacy at the client side requires equipping the client device with sufficient semantics data, while also demanding computational resources to execute the obfuscation algorithm. Therefore, the practicality of the proposed privacy protection mechanism focuses on the spatial overhead of semantic location data and the temporal cost of the location obfuscation algorithm.

Location Semantic Database. The storage space of location semantic database is proportional to the product of location count and semantic vector size. Following in Bordenabe et al. (2014) and Yu et al. (2017), this paper adopts a discrete representation of geographic locations, where more discrete locations correspond to higher representational precision. Our experiments only consider scenarios with no more than 500 locations. In practical applications, higher-precision discrete location representations can be adopted. However, location semantics typically exhibit spatial continuity, meaning that minor location changes do not significantly alter semantic characteristics. Therefore, extremely fine-grained discrete location representations may not yield proportional improvements in semantic accuracy. This paper models location semantics as the distribution of nearby PoIs, thus the content of location semantics depends on the PoI categories used. The evaluation section in this paper considered both one-dimensional and multi-dimensional PoI distribution data. In practical scenarios, the selection of PoI categories as semantic dimensions depends on user preferences. Users can choose to protect only the sensitive PoI categories they care about, or simply include all the feasible PoI categories.

Furthermore, a PoI database covering a specific area can be prestored on the client side. For instance, the Beijing PoI database used in the evaluation section contains approximately 350,000 records and occupies about 38 MB. Given that smartphones typically have storage capacities measured in GB, maintaining such a database on mobile devices does not impose significant burdens. When the privacy preserving mechanism needs to access the semantics of certain locations, it can lazily query the nearby PoI distribution from the database to construct location semantic vectors. The construction of location semantic databases and PoI databases can be delegated to the LBSP. It can prebuild and distribute databases of a specific region (e.g., city-level) to users who need them, thereby saving the overhead of users querying the PoI API (e.g., Overpass API).

Location Obfuscation Mechanism. This paper proposes three location semantic privacy protection algorithms, including Geo-Ind-based methods, OPT-SEM-IND, and heuristic methods. Both Geo-Ind-based and heuristic methods are partially implemented through probabilistic sampling, thus pseudo-locations could be easily obtained even on mobile devices. OPT-SEM-IND differs significantly as it requires solving a linear optimization problem to obtain an obfuscation probability f that satisfies Sem-Ind. The time and computational resources required for solving location obfuscation probabilities primarily depend on the number of locations and Sem-Ind constraints. With the setting adopted in the evaluation section, the solving processes take several minutes, running on a server with 16 CPU cores. It is impractical to place this computational task on mobile devices when generating pseudo-locations, as it will incur substantial delay of LBS request. To address this issue, LBSP can pre-solve the OPT-SEM-IND based on PoI distributions and users' prior distribution to obtain location obfuscation probabilities, which can then be distributed to users, so that users only need to perform probability sampling. As sampling only incurs negligible delay and computation, OPT-SEM-IND can also be applied on mobile devices in practical scenarios.

Overall, from the aspects of semantic database and obfuscation mechanism, all the algorithms proposed in this paper demonstrate feasibility for practical deployment scenarios. Besides, several parameters can be tuned for further practice optimization.

8. Discussion

This paper proposes a new privacy preserving principle called Sem-Ind, aiming at protecting the semantics of locations users visit against leakage. Variants of location obfuscation mechanisms have also been designed and analyzed. However, there are still some problems waiting to be solved. In the case of dealing with multidimensional semantics, the obfuscation of one dimension will incidentally add randomness to the other dimensions which are ignored in this paper. Thus ISS algorithms dealing with multidimensional semantics can result in excessive utility loss. In addition, the greedy manner adopted by NSS and ESSA only considers the closest position and ignores the other ones, which may result in a greater loss of utility.

Furthermore, while most of the content in this paper is derived from a theoretical perspective, practical applications may encounter other issues that are significant and can impact the practicality of the algorithms. The practicality of the privacy preservation mechanisms proposed in this paper in real-world scenarios requires further analysis, including the resources consumed by the interaction protocols between clients and LBSP, as well as the time costs for clients to run the privacy preservation algorithms. Additionally, it is worth further studying whether attackers could exploit other side-channel information to gain more personal private data. These issues are left for us to address in our future work.

9. Conclusion

To protect location semantic privacy, this paper models location semantics through the distributions of PoIs in the vicinity of locations, represented as multidimensional vectors. We introduce Sem-Ind, a novel paradigm for location obfuscation mechanisms, which ensures that the probabilities of any two locations within a region being obfuscated to locations with identical semantics are similar. Sem-Ind shares a comparable structure with LDP-like guarantees, such as Geo-Ind. However, it distinguishes itself by relaxing privacy requirements to prioritize enhanced utility performance relative to Geo-Ind. Three privacy-preserving mechanisms are designed, including Geo-Ind-based methods, OPT-SEM-IND and heuristic methods. Based on defined utility and privacy metrics, these obfuscation mechanisms are empirically evaluated. Experimental results show that OPT-SEM-IND finds the optimal obfuscation function while ensuring privacy. Meanwhile, two heuristic algorithms can quickly perform location obfuscation, at an acceptable level of utility loss.

CRediT authorship contribution statement

Dikai Zou: Writing – review & editing, Writing – original draft, Visualization, Validation, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Jun Tao:** Supervision, Resources, Project administration, Funding acquisition. **Zuyan Wang:** Writing – review & editing, Conceptualization.

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Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

All the code and data is open-sourced and can be reached from https://github.com/Kherrisan/SALPPM.

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